

Volume 15

Cyber Operations Framework (COF)

Version: 1.0

Status: Draft

Vision

Project Sentinel shall become the operating platform for the entire cybersecurity lifecycle. Instead of separate products:

- Pentest
- ASM
- SIEM
- SOAR
- Asset Management
- Compliance
- Vulnerability Management
- Endpoint Security

Everything becomes part of one operational ecosystem.

The Cyber Operations Lifecycle

Every activity follows the same lifecycle.

Discover

Every mission.

Every module.

Every user.

Chapter 1

Mission Workspace

Every mission receives its own workspace.

Example

Mission

Everything related to the mission stays together.

Chapter 2

Digital Mission Folder

Unlike most security products, every mission automatically creates a **Digital Mission Folder**.

It contains:

- Objectives
- Scope
- Permissions
- Timeline
- Evidence
- Findings
- Screenshots
- Reports
- Decisions
- Approvals
- Remediation
- Verification

Nothing is lost.

Everything is searchable.

Chapter 3

Operational Awareness

Mission Control always answers:

Where are we?
What is running?
What changed?
What needs attention?
What is blocked?
What completed successfully?

Chapter 4

Cyber Operations Center

Instead of dashboards.
Think Command Center.
Example

Simple.
Readable.
Professional.

Chapter 5

Mission Map

Every mission automatically builds a visual map.
Company

This becomes the operational picture.

Chapter 6

Mission Timeline

Everything becomes chronological.

09:00

Users can replay everything.

Chapter 7

Mission Health

Instead of only system health.

Mission Health.

Example

Discovery

This immediately tells the operator what is happening.

Chapter 8

Universal Cyber Object

One revolutionary idea.

Everything inherits from one base object.

SentinelObject

One inheritance model.

One API.
One lifecycle.
One identity system.

Chapter 9

Operational State Machine

Every object follows the same lifecycle.
Created

Simple.
Consistent.
Predictable.

Chapter 10

Mission Score

Instead of Scan Score.
Mission Score.
Example
Coverage

Now the user knows whether the mission itself was successful.

PS-ADR-0012

One Operational Model

Decision

All Project Sentinel modules shall follow one operational lifecycle.

Reason

Consistency.

Predictability.

Maintainability.

Scalability.

NEW CORE ENGINE

Cyber Memory Engine (CME)

This is something I have never seen implemented as a first-class capability.

The platform remembers everything.

Not as logs.

As knowledge.

For example.

Six months later.

The platform remembers.

This server has changed ownership three times.

It remembers.

This certificate has expired twice.

It remembers.

This employee repeatedly installs unapproved software.

It remembers.

This website always introduces CSP issues after deployments.

It remembers.

Everything becomes organizational knowledge.

NEW CORE ENGINE

Security Reasoning Engine (SRE)

This is **not AI**.

It is deterministic reasoning.

Example.

Website

Instead of isolated findings.
We build reasoning chains.

NEW CORE ENGINE

Cyber Operations Scheduler

Much more powerful than a cron scheduler.
It understands.
Mission priority.
Business hours.
Maintenance windows.
Dependencies.
Approvals.
Resource utilization.
For example.
Do not scan production during business hours.
Automatically scan staging after deployment.
Verify remediation every Sunday.
Pause cloud scanning during maintenance.
This becomes the platform planner.

NEW IDEA

Sentinel Marketplace

Not an app store.
A **curated security extension ecosystem**.
Organizations can install:

- Discovery plugins
- Compliance packs
- Language packs
- Threat intelligence connectors
- Report templates

- Dashboard layouts
- Visualization modules
- Industry policy packs
- Custom object types
- Workflow templates

Every extension is digitally signed, versioned, permission-scoped, and reviewed before installation. This keeps the platform extensible without compromising security.

NEW IDEA

Sentinel Research Lab

This is aimed at researchers and advanced users.

They can safely develop:

- Detection rules.
- New analysis modules.
- Parsers.
- Visualization techniques.
- Correlation algorithms.
- Plugin prototypes.

The Lab is isolated from production environments and supports experimentation without affecting operational deployments.

NEW IDEA

Cyber Digital Twin

Earlier we introduced Digital Twins for individual assets.

I now propose expanding that concept.

Every organization receives a **Cyber Digital Twin**.

It contains:

- Infrastructure.
- Users.
- Applications.
- APIs.
- Mobile devices.
- Cloud resources.
- Network topology.
- Trust relationships.
- Security posture.
- Historical changes.
- Business criticality.

This is no longer just an inventory.

It is a living model of the organization's cybersecurity environment.

My biggest proposal so far

I believe Project Sentinel should eventually become what I would call a **Cyber Security Operating Environment (CSOE)**.

Not simply software.

An operational environment where every cybersecurity activity takes place.

From a student learning web security...

...to a security consultant performing an authorized assessment...

...to a bank operating a Security Operations Center...

...to a government agency managing critical infrastructure.

The same platform.

The same object model.

The same engineering principles.

Just different missions, permissions, and policy profiles.

Where we are now

At this point, I estimate we have completed approximately **20-25% of the full Blueprint**.

The remaining work is highly structured and includes:

- **Core engines** (Web Security, Endpoint, Mobile, Cloud, Identity, API, Email, Network).
- **Detection architecture** (vulnerability detection, evidence correlation, behavioral analysis).
- **Reporting and documentation.**
- **Plugin SDK and developer platform.**
- **Python implementation architecture** (packages, modules, interfaces, coding standards).
- **Database/storage engine design** (including your custom object engine rather than a traditional relational model).
- **REST and GraphQL APIs.**
- **Desktop and web UI specifications.**
- **Performance and scalability architecture.**
- **Automated testing and QA framework.**
- **Deployment architecture** (Ubuntu, Docker, Kubernetes, air-gapped deployments).
- **Licensing, edition strategy, and governance.**

My recommendation

I think we should continue exactly as we are, but from **Volume 16 onward** every volume should end with four sections:

1. **Architecture Specification** – what we are building.
2. **Python Implementation Plan** – how it will be implemented using Python and open-source technologies.
3. **Testing Strategy** – how we verify that it works correctly and securely.
4. **Future Extension Points** – how the module can evolve without breaking compatibility.

That approach means that when the Blueprint is complete, we won't just have a design document—we will have a **complete engineering specification** that can be implemented systematically, module by module, until Project Sentinel becomes a production-quality Cyber Security Operating Platform.

=====[USER]=====

Good and continue

=====[ASSISTANT]=====

Excellent. I think we are now ready to define what will become the **technical core** of Project Sentinel.

I want Project Sentinel to have something that, to my knowledge, no current cybersecurity platform has implemented as its central architecture:

A Cyber Object Operating System (COOS)

This is **not** an operating system like Windows or Linux.

It is the internal operating model of Project Sentinel.

Everything inside the platform becomes an object with intelligence, relationships, history, permissions, evidence, and lifecycle.

This will become one of our biggest competitive advantages.

PROJECT SENTINEL BLUEPRINT

Volume 16